Predicting Types of Cyber Attacks using Machine Learning Models

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***Abstract*—Cyber attacks pose significant threats to network security, necessitating robust detection and classification mechanisms. This study investigates the application of machine learning for predicting cyber attack types using three diverse datasets: a synthetic cybersecurity attacks dataset, UNSW-NB15, and NSL-KDD. Both binary (normal vs. attack) and multiclass classification tasks were addressed using Logistic Regression, Random Forest, and Histogram-based Gradient Boosting classifiers. A comprehensive preprocessing pipeline was implemented, including data cleaning, missing value imputation, categorical encoding, feature selection, and class imbalance handling via SMOTE (Synthetic Minority Over-sampling Technique) and weighted losses. Experimental results revealed substantial performance variation across datasets. While models achieved high accuracy (up to 0.7991) and macro F1-scores (up to 0.7988) on UNSW-NB15 and NSL-KDD, particularly with Histogram-based Gradient Boosting, performance approached random guessing on the synthetic dataset. These findings highlight the critical influence of dataset realism and quality on model efficacy, demonstrating the superiority of gradient boosting approaches for intrusion detection on authentic network traffic data while underscoring limitations of synthetic datasets in training reliable classifiers.**

***Index Terms*—Cyber attack classification, machine learning, Logistic Regression, Random Forest, Histogram-based Gradient Boosting, UNSW-NB15 dataset, NSL-KDD dataset, class imbalance, feature engineering, binary classification, multiclass classification, network security, anomaly detection.**

I. Introduction

The exponential growth of digital infrastructure has intensified cyber threats, including DDoS attacks, intrusions, and malware, rendering traditional rule-based intrusion detection systems (IDS) inadequate due to their limited adaptability and high false-positive rates. Machine learning provides a data-driven alternative for effective anomaly detection and attack classification.

This study, titled "Predicting Types of Cyber Attacks using Machine Learning Models," evaluates supervised learning approaches on three datasets: a synthetic cybersecurity dataset (DDoS, Intrusion, Malware), the realistic UNSW-NB15 dataset (nine attack categories), and the benchmark NSL-KDD dataset (DoS, Probe, R2L, U2R). Three classifiers—Logistic Regression (baseline), Random Forest (ensemble), and Histogram-based Gradient Boosting (advanced booster)—are compared in binary and multiclass settings, with rigorous preprocessing and class imbalance handling.

This research report aims to evaluate model performance across datasets of varying realism, determine the most effective classifier, and examine the influence of data quality on results. Findings reveal superior performance on realistic datasets and highlight limitations of synthetic data, offering insights for practical machine learning-based IDS deployment.

II. Literature Review

Several studies have applied machine learning techniques to predict different types of cyber attacks by analysing historical network traffic and system log data. A common strategy involves training supervised learning models on labeled datasets containing both normal and malicious activities. Kumar and Singh [1] studied cyber attack classification along with severity estimation using machine learning models such as Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and neural networks. Their results showed that ensemble models achieved higher accuracy in multi-class attack prediction, while deep learning models were better at capturing complex attack patterns.

Similarly, Sri Ambritha and Surendhiran [2] proposed an advanced machine learning framework for cyber attack prediction and prevention. Their work focused on feature selection and classifier optimization to improve prediction accuracy. Decision Trees and Random Forest models demonstrated improved performance with lower false-positive rates, highlighting the importance of proper preprocessing and feature engineering in attack classification tasks.

G. S. et al. [3] extended cyber attack prediction by incorporating attacker behavior analysis along with attack type classification. Using network traffic features, their machine learning models successfully differentiated between multiple attack categories, showing that behavioral patterns play a significant role in predicting cyber attacks.

Hossain et al. [4] introduced the Cyber Attack Detection Model (CADM) based on supervised machine learning algorithms. Statistical feature extraction from network traffic was used to classify attack types, resulting in high classification accuracy. This study confirms that traditional supervised models remain effective when combined with well-defined features.

Comparative performance analysis of machine learning algorithms has also been widely studied. Al-Rashidi [5] compared Random Forest, Logistic Regression, and Neural Networks for cyber attack classification. Random Forest consistently outperformed Logistic Regression for complex attack types, while Neural Networks provided comparable accuracy at the cost of higher computational complexity. These findings support the use of ensemble models in practical intrusion detection systems.

Handling imbalanced datasets is a major challenge in cyber attack prediction. Sarker [5] addressed this issue through adaptive learning and anomaly detection techniques in the CyberLearning framework. The study demonstrated that machine learning models are effective in detecting cyber anomalies and multi-stage attacks, which are difficult to identify using rule-based systems.

Data-driven intrusion detection methods have also shown strong performance. Tauscher et al. [6] proposed a machine learning-based approach that learns attack patterns directly from large-scale network data. Their results showed improved detection of unseen attack types compared to traditional signature-based systems.

Deep learning approaches have been increasingly adopted for cyber intrusion detection. A survey by Vinayakumar et al. [7] reported that CNN and RNN models effectively capture temporal and spatial features in network traffic, leading to improved attack classification. However, challenges such as data imbalance and lack of interpretability remain.

Rekha et al. [8] demonstrated that supervised machine learning techniques significantly outperform conventional intrusion detection methods when sufficient labeled data is available. Additionally, Chowdhury Ripan et al. [8] applied Isolation Forest for anomaly detection, achieving effective classification of cyber anomalies in highly imbalanced datasets.

Despite extensive research, many studies rely on static benchmark datasets and lack real-time adaptability. Moreover, limited attention is given to dataset diversity and generalization. This work aims to address these limitations by applying optimized machine learning models with appropriate preprocessing and evaluation techniques to predict different types of cyber attacks.

A. Summary of Research Gap

Although existing studies demonstrate the effectiveness of machine learning models in cyber attack classification, several gaps remain. Many works focus primarily on detection rather than precise attack type prediction, while others rely on static datasets that may not reflect real-world evolving threats. Additionally, issues such as class imbalance, model interpretability, and real-time deployment are not consistently addressed. These gaps motivate further research into robust, adaptive, and explainable machine learning models for predicting different types of cyber attacks.

III. Methodology

A. Dataset Description

This project employs three publicly available datasets from Kaggle to develop and evaluate machine learning models for network intrusion detection and cyber attack classification. These datasets vary in scale, realism, attack diversity, and complexity, allowing for comprehensive comparative analysis across synthetic, hybrid, and benchmark scenarios.

The first dataset, "Cyber Security Attacks" [9], is a synthetic collection of 40,000 simulated network events spanning 2020–2023. It focuses on multi-class classification of three primary attack types—DDoS, Malware, and Intrusion—with Attack Type as the target variable. Comprising 25 features derived from network traffic logs, alerts, and metadata, it exhibits class imbalance (DDoS predominant) and significant missing values in alert-related columns, necessitating imputation and imbalance handling.

The second dataset, UNSW-NB15 [9], represents a more realistic and contemporary benchmark for intrusion detection. Created by the Australian Centre for Cyber Security Excellence, it contains approximately 2.54 million records (with a widely used subset of 175,341 training and 82,332 testing samples) capturing normal traffic and nine modern attack families (Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms). With 49 engineered features extracted from raw network packets using tools like Bro-IDS and Argus, it supports both binary (normal vs. attack) and multi-class classification. The dataset is characterized by extreme class imbalance, high dimensionality, and minimal missing values, making it ideal for evaluating performance on rare attack detection.

The third dataset, NSL-KDD [10], is an improved version of the classic KDD Cup '99 dataset, designed to eliminate redundancies and artificial difficulty biases. It includes 125,973 training and 22,544 testing samples, each described by 41 features grouped into basic, content, time-based, and host-based categories. The target enables binary classification or five-class categorization (Normal, DoS, Probe, R2L, U2R), with severe class imbalance—particularly for rare U2R and R2L attacks. Although free of significant missing values, its age (based on 1998–1999 traffic simulations) limits the representation of current threats.

Collectively, these datasets provide a progressive spectrum from small-scale synthetic data [9] suitable for initial prototyping, to medium-scale modern hybrid data [9], and established benchmark data [10]. This selection facilitates robust model validation, comparison of performance across varying attack granularities and imbalance levels, and analysis of generalization challenges in cybersecurity applications.

B. Data preprocessing

Data preprocessing was critical to address inconsistencies, noise, and biases inherent in network traffic data. The following steps were uniformly applied across experiments, with adaptations based on dataset-specific needs:

- Data Cleaning: Irrelevant or redundant features (e.g., timestamps in non-temporal analyses) were removed. Outliers in numerical features, such as packet lengths or anomaly scores, were identified using interquartile range methods and capped to prevent skewing.
- Handling Missing Values: Missing entries, prevalent in features like alerts/warnings or proxy information in the first dataset, were imputed using mode for categorical variables and median for numerical ones. In the second and third datasets, minimal missingness was handled similarly to maintain data integrity.
- Encoding Categorical Variables: Categorical features (e.g., protocols, traffic types, attack signatures) were encoded using one-hot encoding for low-cardinality variables and label encoding for high-cardinality ones (e.g., IP addresses) to avoid dimensionality explosion. This ensured compatibility with downstream algorithms.
- Feature Selection and Dimensionality Reduction: Correlation analysis and mutual information scores were used to select top features, reducing multicollinearity (e.g., removing highly correlated traffic metrics in UNSW-NB15). Principal Component Analysis (PCA) was optionally applied in exploratory phases to reduce dimensions from 49 to 20 in the second dataset, preserving over 95% variance.
- Handling Class Imbalance: Significant imbalances were observed, particularly in minority classes like Worms (second dataset) or U2R (third dataset). Synthetic Minority Over-sampling Technique (SMOTE) was employed in multiclass tasks to oversample underrepresented classes, while balanced class weights were integrated into models for binary classification. Undersampling of majority classes was tested but avoided to prevent information loss.
- Feature Engineering and Importance: New features were derived, such as packet rate (bytes per second), from existing timestamps and lengths. Feature importance was assessed using techniques like permutation importance for tree-based models and coefficients for linear ones, highlighting attributes like anomaly scores and packet lengths as pivotal across datasets.

A scikit-learn ColumnTransformer was utilized to orchestrate these steps, combining imputation, encoding, and scaling into a unified pipeline.

C. Learning phase

Three models were implemented for both binary and multiclass classification in each experiment: Logistic Regression (LR), Random Forest (RF), and Histogram-based Gradient Boosting (HGB). These were chosen for their interpretability, robustness to noise, and scalability.

1) Logistic Regression: Selected as a baseline linear model for its simplicity and efficiency in high-dimensional spaces. It models probabilistic outcomes via sigmoid functions, making it suitable for binary tasks. Key hyperparameters included maximum iterations (up to 8000) and L2 regularization (C=1.0) to prevent overfitting.

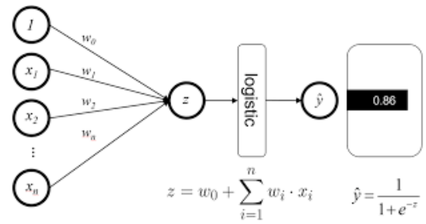

Fig. 1. Logistic Regression Architecture

2) Random Forest: An ensemble method chosen for its ability to handle non-linear relationships and feature interactions through decision tree aggregation. It mitigates overfitting via bootstrapping and random feature subsets. Hyperparameters encompassed a number of estimators (400–600) and maximum depth (none, for full growth), with class weighting for imbalance.

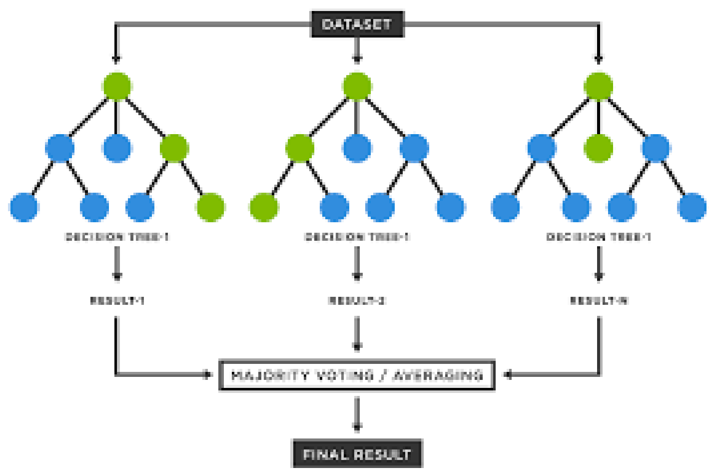

Fig. 2. Random Forest Architecture

3) Histogram-based Gradient Boosting: Adopted for its superior performance on tabular data via gradient descent optimization and histogram binning for efficiency. It excels in handling imbalances and noisy features. Parameters included maximum iterations (400–600), learning rate (0.06–0.08), and early stopping to curb overfitting.

Fig. 3. Histogram-based Gradient Boosting Architecture

Models were implemented using scikit-learn, with pipelines ensuring seamless integration of preprocessing and classification. Datasets were split into 80% training and 20% testing sets using stratified sampling to preserve class distributions. For hyperparameter optimization, 5-fold cross-validation was performed on the training set via GridSearchCV, focusing on macro F1-score to account for imbalances. Models were trained on CPU-optimized environments, with random seeds for reproducibility.

D. Evaluation Metrics

Performance was quantified using accuracy for overall correctness, precision and recall for class-specific error analysis, macro-averaged F1-score for balanced assessment across classes, and Area Under the ROC Curve (AUC) for binary tasks to evaluate discrimination. Confusion matrices and ROC curves were visualized to identify misclassification patterns.

IV. Results and Discussion

A. Quantitative Performance Analysis

Models were evaluated on test sets from each dataset, yielding insights into their efficacy for binary and multiclass classification.

For the first dataset (cybersecurity attacks), binary models achieved accuracies of 0.3301 (LR), 0.323 (RF), and 0.3305 (HGB), with macro F1-scores of 0.33, 0.32, and 0.33, respectively. Multiclass results showed accuracies of 0.3302(LR), 0.323(RF), and 0.3305(HGB), with macro F1-scores of 0.33, 0.32, and 0.33. HGB exhibited the highest AUC (approximately 0.49) in binary tasks, as depicted in ROC curves and the Confusion Matrix (Fig. 4).

On the UNSW-NB15 dataset, binary accuracies were 0.7257 (LR with balanced weights), 0.8183 (RF with balanced subsampling), and 0.8261 (HGB), accompanied by macro F1-scores of 0.7253, 0.8155, and 0.8232. Multiclass performance highlighted challenges with minority classes, as evidenced in classification reports (Table I). Confusion matrices (Fig. 5) revealed frequent misclassifications of Analysis and Worms as Normal.

The NSL-KDD dataset produced binary accuracies of 0.7536 (LR), 0.7711 (RF), and 0.7991 (HGB), with macro F1-scores mirroring those of the first dataset (0.7532–0.7988). Multiclass accuracies were 0.7608 (LR), 0.7511 (RF), and 0.7651 (HGB), with macro F1-scores of 0.5685, 0.4967,

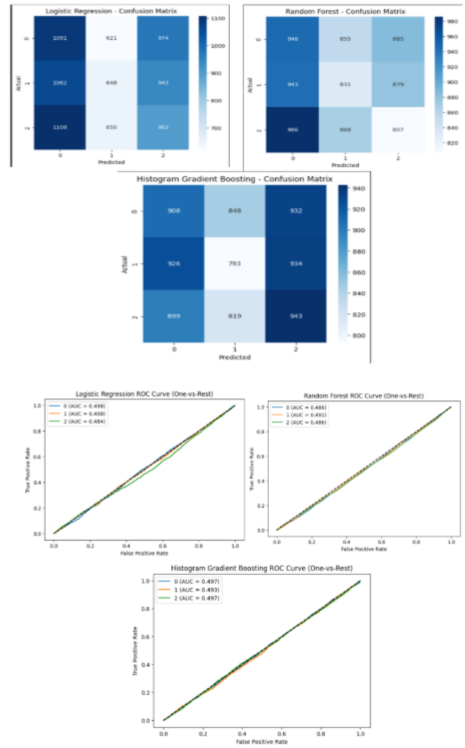

Fig. 4. Confusion Matrix and ROC curves of the Dataset 1 model run

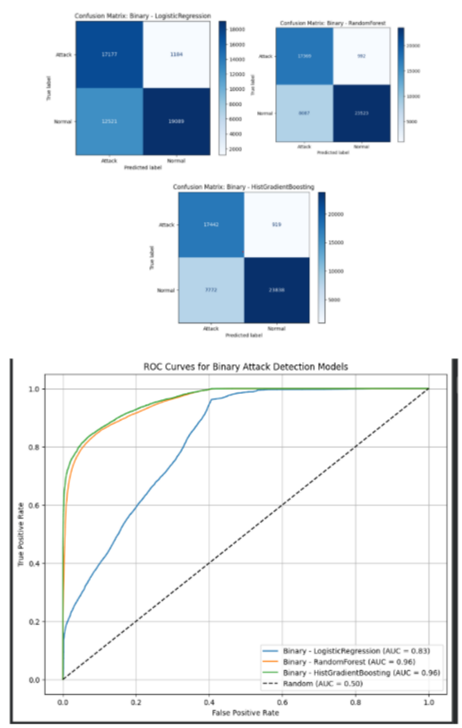

Fig. 5. Confusion Matrix and ROC curves of the Dataset 2 model run

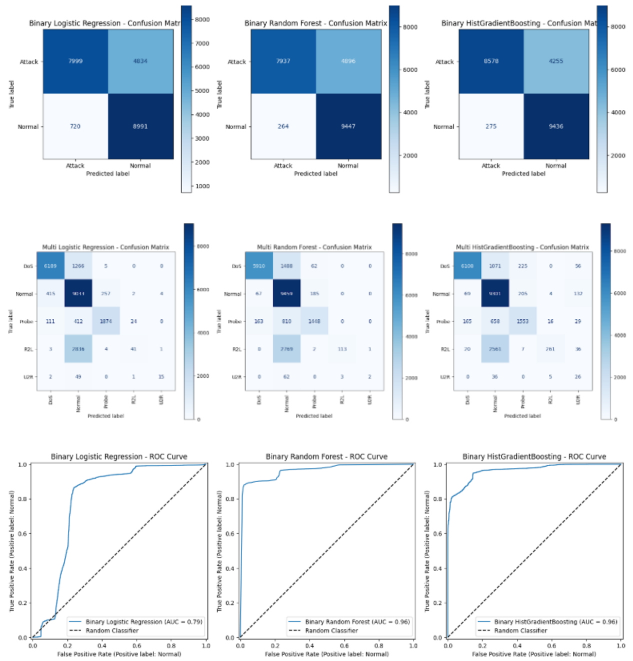

Fig. 6. Confusion Matrix and ROC curves of the Dataset 3 model run

and 0.5400. Feature importance plots (Fig. 6) underscored attributes like anomaly scores in driving predictions.

Table I summarizes key metrics across datasets:

TABLE I
SUMMARY OF KEY METRICS ACROSS DATASETS

Dataset	Model	Binary Accuracy	Binary Macro F1	Multiclass Accuracy	Multiclass Macro F1
3*Cybersecurity Attacks	LR	0.3301	0.33	0.3302	0.33
	RF	0.323	0.32	0.323	0.32
	HGB	0.3305	0.33	0.3305	0.33
3*UNSW-NB15	LR	0.7257	0.7253	0.5224	0.3259
	RF	0.8183	0.8155	0.7592	0.4978
	HGB	0.8261	0.8232	0.7724	0.4581
3*NSL-KDD	LR	0.7536	0.7532	0.7608	0.5685
	RF	0.7711	0.7701	0.7511	0.4967
	HGB	0.7991	0.7988	0.7651	0.5400

B. Comparison of Model Performance

Across datasets, HGB demonstrated consistent superiority in binary classification on the UNSW-NB15 and NSL-KDD datasets, achieving accuracies up to 0.8261 and macro F1-scores up to 0.7991, owing to its effective handling of noisy features through gradient boosting. However, on the first dataset, all models performed near random chance (accuracies 0.33), with negligible differences among LR, RF, and HGB. In multiclass scenarios, HGB retained an edge on UNSW-NB15 and NSL-KDD (e.g., 0.7724 accuracy on UNSW-NB15), but macro F1-scores remained below 0.57, primarily due to low recall for rare classes such as U2R (0.22–0.39). RF showed strong precision for majority classes (e.g., 0.97 for Attacks in binary tasks on NSL-KDD), whereas LR provided a stable baseline but faltered under imbalances, as indicated by reduced F1-scores (e.g., 0.3259 on UNSW-NB15 and 0.33 on the first dataset).

Comparative AUC values ranged from 0.49 (near random) on the first dataset to 0.77–0.83 on others, with ROC curves (Fig. 1) illustrating HGB's proximity to optimal performance on realistic datasets. Variations in performance were linked to dataset characteristics: the larger, more realistic UNSW-NB15 exposed imbalance vulnerabilities yet supported higher accuracies, while NSL-KDD's curated structure promoted better generalization. In contrast, the first dataset's synthetic nature resulted in uniformly poor outcomes across models.

C. Final Comparative Analysis

Despite variable performance, the Histogram-based Gradient Boosting model stood out as the overall best performer, delivering the highest accuracies (0.7651–0.7991) and macro F1-scores (0.4581–0.7988) on the UNSW-NB15 and NSL-KDD datasets, substantiated by its resilience to noise and imbalances. However, its near-random efficacy on the first dataset underscores limitations in synthetic environments. UNSW-NB15 emerged as the most reliable dataset, owing to its scale, realism, and ability to facilitate thorough evaluations, even amid challenges. These results affirm gradient boosting's value for intrusion detection on authentic data, while advocating for enhanced feature engineering or hybrid models in future research to overcome synthetic data deficiencies and minority class biases.

V. Conclusion

This study evaluated the performance of Logistic Regression, Random Forest, and Histogram-based Gradient Boosting models for predicting cyber attack types across three datasets of varying realism. Results consistently demonstrated that Histogram-based Gradient Boosting outperformed the other classifiers on realistic datasets (UNSW-NB15 and NSL-KDD), achieving accuracies up to 0.8261 and macro F1-scores up to 0.7988 in binary classification, owing to its robustness against noise and class imbalances. In contrast, all models exhibited near-random performance on the synthetic cybersecurity dataset, highlighting the critical limitations of artificially generated data lacking discriminative patterns.

The findings underscore the paramount importance of dataset quality and realism in training effective intrusion detection systems. While imbalance mitigation techniques improved minority class recall, challenges persisted in multiclass settings, particularly for rare attack types. Overall, Histogram-based Gradient Boosting emerged as the most effective model for practical deployment on authentic network traffic, providing a strong foundation for future machine learning-based IDS. Future work should focus on advanced feature engineering, hybrid ensembles, and real-time evaluation to further enhance detection of evolving cyber threats.

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